Lab 2 - Knowledge Is Power (KIP)

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to explain what the Knowledge Is Power (KIP) prototype must do. It describes the main features, users, and requirements of the system. Team Crystal can use this document as a guide while building and testing KIP.

1.2 Scope

KIP is an interactive web application that teaches practical legal information to young people, mainly learners between the ages of 16 and 19. Many young people must follow laws that they may not fully understand. KIP helps by using simple, game-based legal scenarios. Users make choices and receive immediate feedback.
The prototype focuses on Virginia traffic law and copyright law. Users can browse games, filter them by topic, view game statistics, play scenarios, and review completed games. KIP also saves user progress and uses it to recommend areas that need improvement. KIP is an educational tool and does not replace legal advice.

1.3 Definitions, Acronyms, and Abbreviations

The following entries are some of the unique acronyms that are used in this document.

CDC: Centers for Disease Control and Prevention.

KIP: Knowledge Is Power.

MFCD: Major Functional Component Diagram.

UX: User Experience

1.4 References

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1.5 Overview

Section 2 gives an overall description of KIP. It explains the product, main functions, users, constraints, and assumptions. Section 3 is the group portion and lists the detailed system requirements.

2 Overall Description

2.1 Product Perspective

KIP is a browser-based legal learning platform. It combines educational content, game scenarios, progress tracking, and immediate feedback. The current application is a prototype of a larger real-world product. The front end uses Angular, HTML, and Bootstrap. It includes the Dashboard, Browser, Game Window, Progress, Settings, and administration pages. The back end uses Java and Spring Boot to manage accounts, games, feedback, and progress. MySQL stores the application data. JHipster provides the project structure, and Liquibase manages database updates and starting data.

2.2 Product Functions

The KIP prototype provides the following principal functions:

- **Navbar:** Maintains consistent navigation to the Dashboard, Games, Progress, and Settings pages.
- **Dashboard:** Displays a welcome message, daily recommendation, profile summary, and recent activity.
- **Browser:** Lists available games, presents game details, filters games by legal topic, and displays progress-based game statistics.
- **Game Window:** Presents the active scenario, evidence, response choices, and scenario progress to the learner.
- **Admin Panel:** Provides authenticated administrative access to supported platform settings.
- **User Settings:** Allows a learner to configure supported account and interface settings.
- **Progress Tracker:** Stores concept, topic, and game progress and displays the learner's completed-game history.

- **Feedback System:** Records responses and returns explanatory feedback for questions and completed scenarios.
- **Legal Reference Algorithm:** Checks legal-source information and identifies changes that may affect stored legal content.
- **Game Engine:** Loads games, manages active sessions, processes responses, advances stages, and completes or abandons sessions.
- **Curation System:** Uses progress information to recommend topics or concepts that may need additional practice.

2.3 User Characteristics

2.3.1 Visitor

A visitor is a person who is not signed in. A visitor can view public information and the sign-in or registration options. The visitor does not need technical knowledge.

2.3.2 Learner

A learner is the main user of KIP. The learner is usually a teenager or young adult who wants to understand basic legal topics. Learners may not know legal terms, so KIP must use clear language and simple navigation. A signed-in learner can browse and filter games, play scenarios, view statistics, and review progress.

2.3.3 Administrator

An administrator is a signed-in user with permission to use administrative tools and platform settings. The administrator manages supported system options that regular learners cannot access.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A